John Carl Pogue Planetarium — Grade 3 Space Games

Work in a group of up to 3. Open the planetarium games website and click **Grade 3**. Do each game, then circle your answers.

Group (up to 3): 1. ____________________ 2. _____________________ 3. _______________________

Teacher: _______________________________ **Date:** _____________________

Game 1 — Day & Night

How to play:

1. Open the game. You will see the Earth and the Sun. Find the box that says **Clock in Austin**. It reads 4:00 P.M.
2. Press the yellow **Spin 12 hours** button. Watch the Earth spin.
3. Read the three boxes: the clock, **Austin right now** (day or night), and **Earth has turned**.
4. You can also slide the bar to spin the Earth to other times.

Circle one letter for each question (read the answer choices on the screen):

1. During the 12 hours after 4:00 P.M., which event will happen?

A B C D

2. It is 2:00 A.M. and the sky is dark. What is the Sun doing right now?

A B C D

Game 2 — One Day, One Year

How to play:

1. Open the game. The Sun is in the middle and the Earth is on the dotted path.
2. Press **Spin once (1 day)**. Watch the Earth spin one time. Read the words that appear.
3. Press **Trip around the Sun (1 year)**. Watch the Earth go all the way around. Read the words.
4. Press **Reset** to try again.

Circle one letter for each question:

1. Which of these takes Earth about 24 hours?

A B C D

2. About how long does it take Earth to travel halfway around the Sun?

A B C D

3. Why does the Sun appear to rise in the east and set in the west every day?

A B C D

Game 3 — Sun & Shadows

How to play:

1. Open the game. You will see a tree and the Sun in the sky.
2. Slide the bar from **Morning** to **Evening**, or press **Play the day**.
3. Watch the Sun move across the sky and the shadow change — long, then short, then long the other way.
4. Read the sentence under the picture (morning, midday, or evening).

Circle one letter for each question:

1. Which sentence describes a result of the rotation of the Earth on its axis?

   A B C D

2. The Sun looks like it moves across the sky. What is really moving?

   A B C

3. Why does the Sun appear to rise in the east and set in the west every day?

   A B C D

Game 4 — Long Days, Short Days

How to play:

1. Open the game. Press **Measure again** and watch the two bars fill.
2. Read the hours: **Chicago ≈ 9 hours** (short), **Sydney ≈ 14 hours** (long).
3. Press **Lean Sydney to the Sun**, then **Lean Chicago to the Sun**. Read the words about summer and winter.

Circle one letter for each question:

1. Sydney has about 14 hours of daylight. Which season is Sydney most likely having?

   A B C

2. Chicago has only about 9 hours of daylight. Which season is Chicago most likely having?

   A B C

3. Which clue best shows that Sydney has long days?

   A B C

Game 5 — Summer & Winter Days

How to play:

1. Open the game. Press ☀ **Summer**. Watch the Sun make a long, high path. Read **Hours of daylight** (about 14).
2. Press ❄ **Winter**. Watch the Sun make a short, low path. Read the hours (about 10).
3. Press **Play the day** to move the Sun across the sky.

Circle one letter for each question:

1. Why are the daytime hours longer in summer than in winter?

   A B C D

2. In winter, the Sun takes a ______ path, so there is ______ daylight.

   A B C D

3. A town has about 14 hours of daylight. Which season is it most likely?

   A B C

Game 6 — Moon Phases

How to play:

1. Open the game. **January 1 is the new moon.**
2. Drag each moon from the left onto the day you think it happens: **First quarter**, **Full moon**, **Third quarter**, **New moon**.
3. If a moon is in the wrong spot, read the hint and try again. Remember: new moon to new moon is about 29.5 days, and the full moon is halfway.

Circle one letter for each question:

1. About 15 days after the new moon (halfway), which phase does the Moon show?

   A B C D

2. A new moon is on January 30. On about which day in February will the next new moon be?

   A B C D